Lecture 6 (February 14)

Ilani's office hours have been voted to be Friday at 3PM in 2-390c.

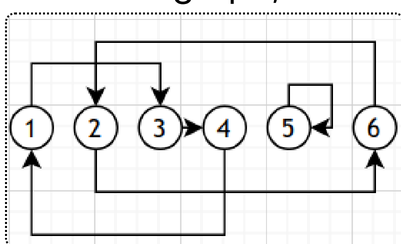
Permutations

Recall that permutations w are elements of the symmetric group S_n. There are a few different notations for permutations:

- 1-line notation: $w = w_1, \dots, w_n$. For example, $w = 3, 6, 4, 1, 5, 2$.
- 2-line notation: sometimes it's more convenient to use two lines, like this:

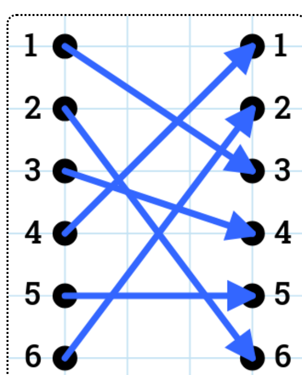
$$w = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 6 & 4 & 1 & 5 & 2 \end{pmatrix}$$

- Graphical notation: Draw a directed graph, with an edge from i to w_i.



- Cycle notation: Note how our graph has three cycles. We record these cycles in order, like $w = (1, 3, 4)(2, 6)(5)$. (We call 5 a fixed point of the permutation.)

We denote by $[n]$ the set $\{1, 2, 3, \dots, n\}$. Note that a permutation is essentially also a bijection from $[n]$ to itself (namely the one that sends i to w_i). So we can also represent our permutation like this:



We can also represent permutations using permutation matrices A_w, where column i consists of 0 everywhere except w_i (which is a 1). For example,

$$A_w = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Multiplying permutations

Suppose we have two permutations $u, w \in S_n$. We can think of their product uw as either the composition of the two maps $i \mapsto u(w(i))$, or the product of their permutation matrices $A_u A_w$.

Definition (Adjacent transpositions).

The **adjacent transpositions** s_i are small permutations made by swapping adjacent elements, i.e. $(i, i+1)$ in cycle notation.

What happens when you multiply adjacent transpositions? Let's find $s_1 \cdot s_2$ in S_3.

$$s_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$
$$s_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

We can multiply them by first applying s_2 and then applying s_1. For example, $s_1 s_2(3) = s_1(s_2(3)) = s_1(2) = 1$.

$$s_1 s_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

Statistics on permutations

A **statistic** σ is a function from S_n to $\mathbb{Z}_{\geq 0}$. Given a statistic, we can write its **generating function** as

$$f_{\sigma,n}(x) = \sum_{w \in S_n} x^{\sigma(w)}.$$

Two statistics σ, μ are **equidistributed** if they have the same generating function $f_{\sigma,n}(x) = f_{\mu,n}(x)$ (i.e. every output appears with the same frequency).

Some frequently-used examples of statistics:

- Number of inversions:

$$\mathrm{inv}(w) := \#\{(i,j) | \ 1 \le i < j \le n \quad \text{and} \quad w_i > w_j\}$$

- Number of descents:

$$\mathrm{des}(w) := \#\{i \in [n-1] \mid w_i > w_{i+1}\}$$

- Number of cycles:

$$\mathrm{cyc}(w) := \text{number of cycles in } w \text{ (including fixed points)}$$

- Major index (named for Major MacMahon):

$$\mathrm{maj}(w) := \sum_{w_i > w_{i+1}} i$$

- Number of records (also known as left-to-right maxima):

$$\mathrm{rec}(w) = \#\{i \mid w_i > w_j \quad \forall j < i\}$$

- Number of exceedances:

$$\mathrm{exc}(w) = \#\{i \mid w_i > i\}$$

- Number of anti-exceedances:

$$\mathrm{aexc}(w) = \#\{i \mid w_i < i\}$$

- Number of weak exceedances:

$$\mathrm{wexc}(w) = \#\{i \mid w_i \ge i\}$$

  Different representations of permutations can help for computing statistics. For our example $w = 3, 6, 4, 1, 5, 2$:

$$
\begin{aligned}
\operatorname{inv}(w) &= 9 && \text{(count intersections in bipartite graph)}\\
\operatorname{des}(w) &= 3 && \text{(use one-line notation)}\\
\operatorname{cyc}(w) &= 3 && \text{(use cycle notation)}\\
\operatorname{maj}(w) = 2+3+5 &= 10 && \text{(use two-line notation)}\\
\operatorname{rec}(w) &= 2 && \text{(use one-line notation)}\\
\operatorname{exc}(w) &= 3 && \text{(use two-line notation)}\\
\operatorname{aexc}(w) &= 2 && \text{(use two-line notation)}\\
\operatorname{wexc}(w) &= 4 && \text{(use two-line notation)}
\end{aligned}
$$

There are a few important classes of equidistributed statistics.

- Statistics equidistributed with $\operatorname{des}(w)$ are called **Eulerian statistics**, because its generating function is the Eulerian polynomial (discussed later)
- **Mahonian statistics** are equidistributed with $\operatorname{maj}(w)$ (also named for MacMahon)
- **"Stirlingian statistics"** (not a canonical name) are equidistributed with $\operatorname{cyc}(w)$.

In fact, all of the above statistics fall into one of these categories!

Theorem 1 (Equidistributed statistics).

Using the notation $\sim$ for equidistribution:

1. $\operatorname{des} \sim \operatorname{exc} \sim \operatorname{aexc} \sim (\operatorname{wexc} - 1)$
2. $\operatorname{inv} \sim \operatorname{maj}$
3. $\operatorname{cyc} \sim \operatorname{rec}$

We want to find interesting bijections that turn these statistics into each other. For example, here is a proof of $\operatorname{cyc} \sim \operatorname{rec}$.

Proof. We want to find a bijection between permutations with k cycles and those with k records.

Note that permutations can be written in cycle notation in different ways (as you can change the starting point of each cycle and rearrange them). For example, $w = (1, 3, 4)(2, 6)(5) = (2, 6)(3, 4, 1)(5)$.

Let's specifically write them so that the starting point a_i of each cycle is its maximal element, and sort the cycles in increasing order of maximal element. In this case, we'd have $w = (4, 1, 3)(5)(6, 2)$. Our map is to just erase the parentheses reinterpret this as one-line notation: $\tilde{w} = 4, 1, 3, 5, 6, 2$.

We make two claims that complete the proof.

1. $\mathrm{cyc}(w) = \mathrm{rec}(\tilde{w})$. The records of $\tilde{w}$ are exactly the maximal elements of the cycles of w.
2. The map $w \mapsto \tilde{w}$ is a bijection. It suffices to find a reverse map, which is simply starting a new cycle at each record (by placing parentheses before them).

Proving the other equidistribution relations is left to the students.

Lecture 7 (February 18)

Prof. Postnikov had something come up so Ilani (our TA) is filling in today. We will discuss the Robinson-Schensted correspondence that relates permutations and standard Young tableaux.

Robinson–Schensted correspondence

Recall that f_λ represents the number of standard Young tableaux of shape λ.

Theorem 2 (RSK).

There is a bijection

$$S_n \leftrightarrow \{(P, Q) \mid P, Q \text{ are SYTs of the same shape } \lambda \vdash n\}$$

Note that the shape λ can vary, but must be the same for the two tableaux in each pair.

This has a generalization to "semi-standard Young tableaux", the Robinson–Schensted–Knuth correspondence (RSK). We will refer to this theorem as RSK even though it's technically a special case.

In order to start the bijection, we define an "insertion algorithm".

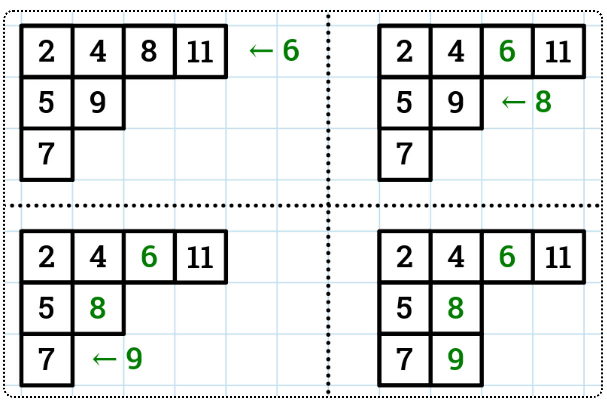
Definition 3 (Schensted insertion).

Suppose we have an "incomplete standard tableau" (a Young diagram filled with numbers that increase in rows and columns, but where some may be missing) and we want to insert a missing number i. **Schensted insertion** is a process that lets us do this.

The steps are as follows.

1. If i is bigger than everything else in the first row, place it at the end.
2. If not, it replaces the first number larger than it, which is then inserted into next row.
3. This process repeats until some number is placed at the end of a row.

An example can be seen below:



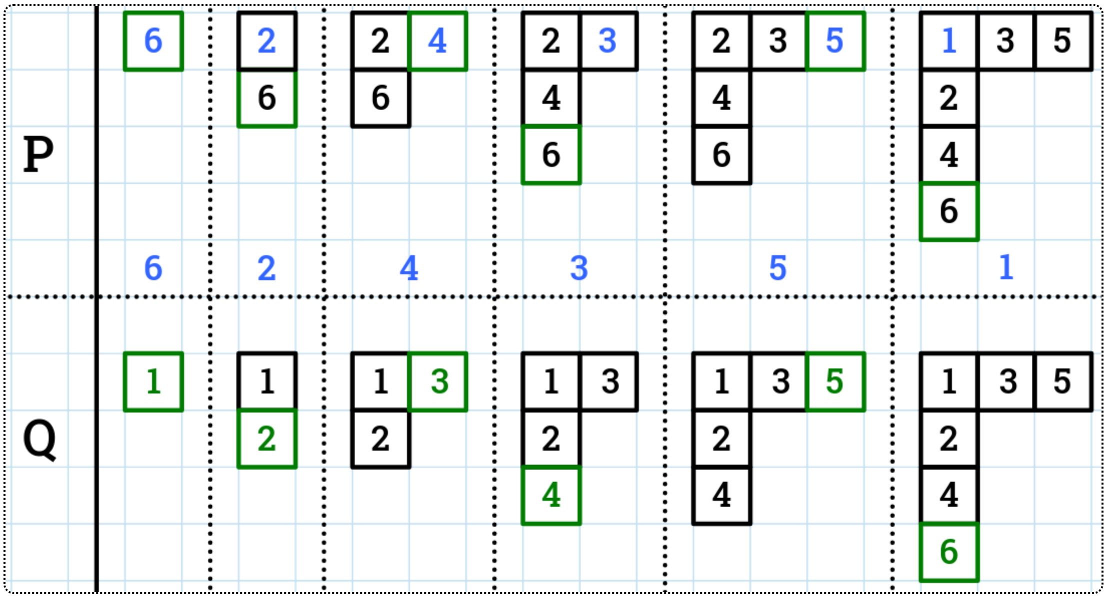
One can confirm that this process preserves the property that rows and columns increase.

The RSK algorithm, mapping a permutation $w = w_1, w_2, \ldots, w_n$ to a pair (P, Q) of Young tableaux, is as follows:

- Start with $P = Q = \varnothing$.
- Schensted insert $w_1, w_2, \ldots, w_n$ in order into P.

- When inserting w_i into P, a box was created somewhere. Add a box in the same place to Q with entry i. (Note that this is not necessarily the same place where w_i went.)

Here's how it looks for the permutation $w = 6, 2, 4, 3, 5, 1$:



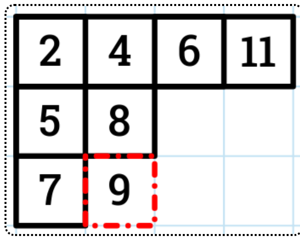
The inserted items (in permutation order) are shown in blue, while the new box added at each step is shown in green.

Note that in every step of our process, we preserve that P and Q have the same shape (since we add boxes in the same location). So, it suffices to show that there reverse the process. We can essentially just reverse each step at the time: we can figure out which box of P is new based on where the largest number in Q is.

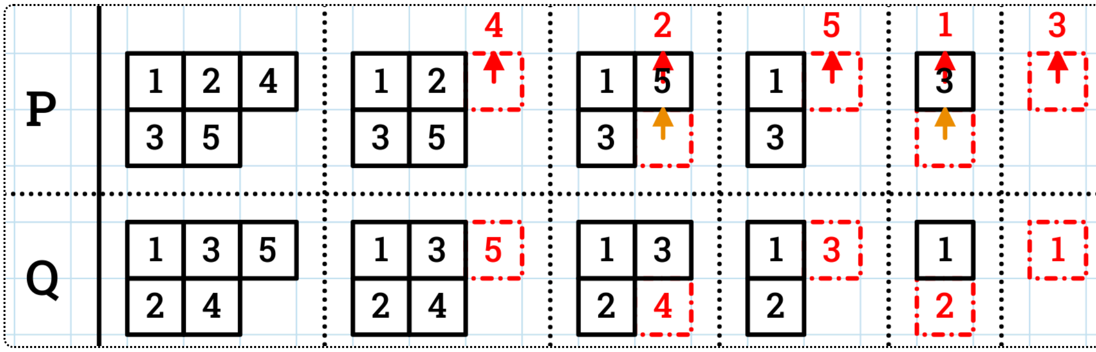
Then, let's see what the process of "un-inserting" a box from P does:

- If we're removing k from the rth row of P, then k must have originally been in the $(r-1)$th row. It would've been knocked out by some number m.
- m must be the currently-largest number in the $(r-1)$th row that is less than k (since otherwise it would have knocked out some other number).
- If $r-1$ is not the first row, m must have also been knocked out by another number, so we repeat until we figure out what the original inserted number was.

In our initial example (seen below), if we know the red box was the most recently added, we can determine that 9 was displaced by 8, which was displaced by 6, so the last insertion was of 6.



This lets us completely reverse-engineer what our original permutation was, using P and Q. Here's an example.



We un-inserted $4, 2, 5, 1, 3$ in order, so our original w must have been the reverse: $3, 1, 5, 2, 4$, which does indeed yield our P and Q under RSK.

Note that multiple permutations may yield the same P tableau, so we need the Q tableau to know where to uninsert from during each step. (Look at e.g. $w = 2, 1, 3$ and $w = 2, 3, 1$, for example.)

A few corollaries naturally follow from RSK, as follows:

Corollary (Sum of f_λ^2).

By counting the number of elements in each set of our bijection, we have

$$\sum_{\lambda \vdash n} (f_\lambda)^2 = n!.$$

Recall that f_λ is the number of SYTs for a shape λ.

Also, using the following theorem (which we will not prove),

Theorem 4 (Inverting permutations).

If running RSK on $w \in S_n$ yields (P, Q), then w^{-1} yields (Q, P).

we have the following:

Corollary (Sum of f_λ).

$$\sum_{\lambda \vdash n} f_\lambda = \#\{(P, P) \mid P \text{ a SYT of shape } \lambda \vdash n\} = \#\{w \in S_n \mid w = w^{-1}\},$$

i.e. the number of involutions in S_n.

Greene's Theorem

Given a permutation $w \in S_n$, we call the shared shape λ of P and Q (after running RSK) the **Schensted shape** of w. For example, the Schensted shape of $3, 1, 5, 2, 4$ is $(3, 2)$, as seen above.

We will also need to define increasing/decreasing sequences within a permutation. An **increasing sequence** of $w_1, w_2, \ldots, w_n$ is a subsequence $w_{i_1} < w_{i_2} < w_{i_3} < \ldots w_{i_k}$ where $i_1 < i_2 < i_3 < \cdots < i_k$. The longest increasing sequence in $3, 1, 5, 2, 4$ is $1, 2, 4$. Decreasing sequences are defined similarly.

Theorem 5 ((part of) Greene's theorem).

Let λ be the Schensted shape of w. Then:

- the size λ_1 of the first row in λ is the length of the longest increasing sequence in w, and
- the size λ_1' of the first column in λ is the length of the longest decreasing sequence in w.

Greene's theorem can also tell us about the total size of the first i columns/rows for larger i, but it requires you to consider the unions of disjoint sequences and we won't go into it right now.

The standard proof of Greene's theorem uses a lot of machinery, but for this version, it may help to consider the locations where each number in a maximally increasing/decreasing subsequence is initially inserted (even if it gets displaced afterwards).

Application

A few lectures ago we said that queue-sortable permutations, i.e. $3\,2\,1$-avoiding permutations, were counted by the Catalan numbers. We can prove this now!

Let $?_n$ be the number of $3\,2\,1$-avoiding permutations of size n.

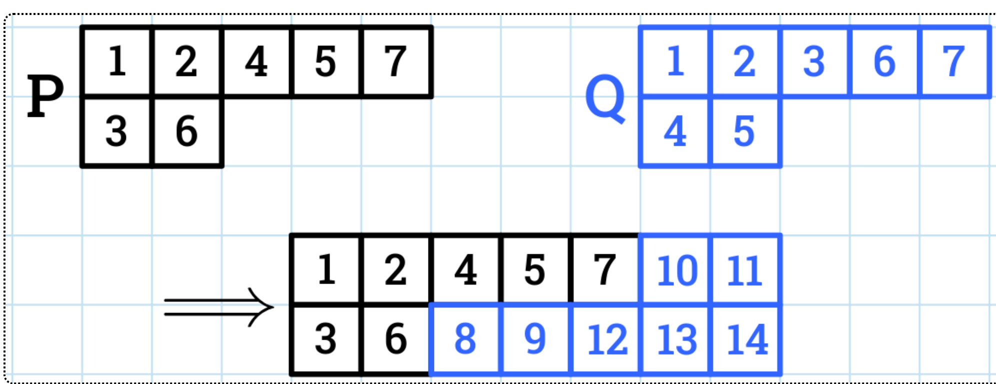
- A permutation is $3\,2\,1$-avoiding if and only if it has no decreasing sequence of length 3;
- in other words, its Schensted shape has at most 2 rows;
- so $?_n$ is equal to the number of pairs (P, Q) of any shape $\lambda \vdash n$ with at most 2 rows.

Recall that there are C_n SYTs of shape (n, n) (via bijection to Dyck paths). So, can we biject between pairs (P, Q) and SYTs of shape (n, n)?

Yes! The forwards operation is as follows:

- Rotate Q by 180°, and replace each entry i with $2n + 1 - i$.
- Glue it onto P.

For example:



To recover P and Q from a SYT of shape (n, n), note that P is just the subtableau formed by the numbers $1, 2, \ldots, n$, and we can recover Q by reversing our steps.

Lecture 8 (February 19)

Today we will introduce Eulerian numbers.

Eulerian numbers (1755)

Euler figured out lots of things, like the infinite sum

$$f_0(x) = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}.$$

Let's make it slightly more complex:

$$f_1(x) = 1 + 2x + 3x^2 + 4x^3 + \cdots = \ ?$$

This is the derivative of our first sum times x, so we can figure out that it's $\frac{1}{(1-x)^2}$.

How about

$$f_2(x) = 1 + 2^2x + 3^2x^2 + 4^2x^3 + \cdots = \ ?$$

In general, these are infinite power series defined by

$$f_n(x) = \sum_{k \geq 0} (k+1)^n x^k.$$

We have the recurrence relation $f_n(x) = (xf_{n-1}(x))'$. For example,

$$f_2(x) = \left(\frac{x}{(1-x)^2}\right)' = \frac{(1-x)^2 + 2x(1-x)}{(1-x)^4} = \frac{1+x}{(1-x)^3}.$$

Then

$$
\begin{aligned}
f_3(x) &= \left(\frac{x(1+x)}{(1-x)^3}\right)' \\
&= \frac{(1+2x)(1-x)^3 + 3(x+x^2)(1-x)^2}{(1-x)^6} \\
&= \frac{(1+2x)(1-x) + 3(x+x^2)}{(1-x)^4} \\
&= \frac{1+4x+x^2}{(1-x)^4}.
\end{aligned}
$$

Euler computed a bunch of these.

$f_1(x)$	$\frac{1}{1-x}$
$f_2(x)$	$\frac{1}{(1-x)^2}$
$f_3(x)$	$\frac{1+x}{(1-x)^3}$
$f_4(x)$	$\frac{1+4x+x^2}{(1-x)^4}$
$f_5(x)$	$\frac{1+11x+11x^2+x^3}{(1-x)^5}$
$f_6(x)$	$\frac{1+26x+66x^2+26x^3+x^4}{(1-x)^6}$

Note that these are all of the form $\frac{A_n(x)}{(1-x)^{n+1}}$ for some polynomial $A_n(x)$. These polynomials have some increasingly interesting properties:

- $A_n(x)$ has degree $n-1$
- $A_n(x)$ has positive integer coefficients
- $A_n(x)$ is palindromic (the x^i coefficient equals the x^{n-1-k} coefficient)
- The sum of the coefficients of $A_n(x)$ is $n!$
  Maybe we can find a combinatorial interpretation of these coefficients, especially since they add to $n!$ (the number of permutations in S_n). In particular, maybe we can find some statistic of permutations that has $A_n(x)$ as its generating function.

We define the **Eulerian number** $A(n,k)$ as the x^k coefficient of $A_n(x)$ (known as the **Eulerian polynomials**). More formally,

$$A_n(x) = \sum_{k=0}^{n-1} A(n,k) \cdot x^k.$$

Remark (Terminology).

Be careful not to confuse Eulerian numbers with Euler numbers, which we will actually discuss later in class, or Euler's number e. Euler did a lot of work with numbers!

Theorem 6 (Eulerian statistics).

$A(n, k)$ is:

- the number of permutations $w \in S_n$ with exactly k descents,
- and also the number of permutations $w \in S_n$ with exactly k exceedances.

Recall that a descent is when $w_i > w_{i+1}$ for some $i \in 1, 2, \dots, n-1$, and an exceedance is when $w_i > i$ for some $i \in 1, 2, \dots, n$.

Note that this implies all of our properties that we noticed above: the sum of coefficients will just be the total number of permutations S_n, they will be positive integers, and they will be palindromic (since $\tilde{w} = w_n w_{n-1} \dots w_1$ has $n - 1 - \mathrm{des}(w)$ descents.)

Proof. First we will show that $\mathrm{des} \sim \mathrm{exc}$ (recall that $\sim$ denotes equidistribution). This was a part of Theorem 1 from last week.

Last time (in Lecture 6), in order to prove that $\mathrm{cyc} \sim \mathrm{rec}$, we constructed bijections between permutations with k cycles and those with k records. We did this by writing w in cycle notation, shifting the cycles to start with maximal entries, reordering the cycles in increasing order of first entry, and reinterpreting it as one-line notation.

$$w = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 5 & 7 & 3 & 1 & 6 & 8 & 4 \end{pmatrix} = (1,2,5)(3,7,8,4)(6) = (5,1,2)(6)(8,4,3,7)$$
$$\implies \tilde{w} = \boxed{5}, 1, 2, \boxed{6}, \boxed{8}, 4, 3, 7$$

I claim that this exact bijection also proves that $\mathrm{des} \sim \mathrm{exc}$:

$$\operatorname{des}(\tilde{w}) = \operatorname{aexc}(w) = \operatorname{exc}(w^{-1})$$

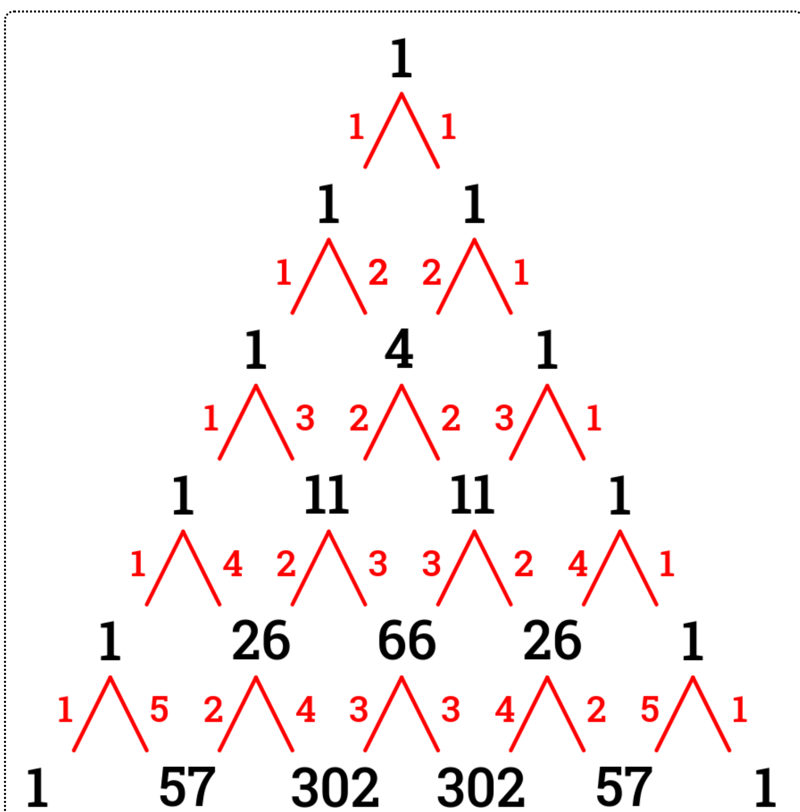
(where aexc denotes anti-exceedances, when $w_i < i$).

For example, in the above, the descents of $\tilde{w}$ are $5 \to 1, 8 \to 4, 4 \to 3$, and the anti-exceedances of w are $3 < 4$, $1 < 5$, $4 < 8$. (Note that these are the same inequalities!) A descent within $\tilde{w}$ occurs exactly when some i is sent to a smaller w_i in w (and we can't create new descents between cycles, since we arranged them in increasing order of maximal element.)

Now let's connect these statistics to $A(n, k)$. Note that we can arrange the Eulerian numbers into an "Eulerian triangle", similar to Pascal's triangle:

$$\begin{array}{ccccccc} & & & A(1,0) & & & \\ & & A(2,0) & & A(2,1) & & \\ & A(3,0) & & A(3,1) & & A(3,2) & \\ \iddots & & & \vdots & & & \ddots \end{array}$$

We can find a recurrence relation similar to Pascal's triangle, where each number is a weighted sum of the two numbers above it based on their k values.



Specifically,

$$A(n,k) = \underline{(n-k)} \cdot A(n-1,k-1) + \underline{(k+1)} \cdot A(n-1,k)$$

for $n \geq 1$, and $A(n,k) = 0$ if $k < 0$ or $k \geq n$.

The following claims finish the proof:

Claim 7 (Recurrence relation and Eulerian polynomials).

The above recurrence relation is true for the definition where $A(n,k)$ is a coefficient of an Eulerian polynomial, since it is equivalent to our original recurrence $f_n(x) = (xf_{n-1}(x))'$.

Claim 8 (Recurrence relation and descents).

The recurrence relation is also "pretty easy to see" if you interpret $A(n,k)$ as the number of permutations in S_n with k descents, and we will prove it next class.

(Hint: Where can you insert n into a permutation of $1, 2, \ldots, n-1$, and what does that do to the number of descents?)

Lecture 9 (February 21)

Problem Set 1 has been posted - please start working on it sooner rather than later! Feel free to ask Prof. Postnikov if you need any help, hints, etc.

- If something is proved in lecture any time before the due date, you can use it, but if it's just mentioned you have to prove it yourself.
- Consulting sources (e.g. textbooks) on problem sets is allowed but you can't say "this problem is solved on page so-and-so in so-and-so book" - you must understand the proof and then explain it in your own words. (Of course, you must cite your sources.) This also applies to using AI (but it's better if you try to solve the problems yourself).
- Collaboration is allowed (but please acknowledge your partners).
- Hand-writing and typesetting are both okay (and in particular writing solutions by hand may make drawing diagrams easier).

Eulerian numbers (continued)

Recall that our coefficient definition of $A(n,k)$ is equivalent to the recurrence relation

$$A(n,k) = (n-k)A(n-1,k-1) + (k-1)A(n-1,k)$$

with $A(1,0) = 1$ and $A(n,k) = 0$ if $k < 0$ or $k \geq n$, by expanding the recurrence relation $f_n = (xf_{n-1}(x))'$. Let's now relate this recurrence to descents in permutations.

Define

$$\tilde{A}(n,k) := \#\{w \in S_n \mid \operatorname{des}(w) = k\}.$$

We shall prove that $A(n,k) = \tilde{A}(n,k)$, by proving that $\tilde{A}$ satisfies the recurrence relation and initial conditions above.

Proof. We obtain permutations $w \in S_n$ from permutations $u \in S_{n-1}$ by inserting the new entry n somewhere in one-line notation.

There are n possible slots: before u_1, between u_1 and u_2, ..., between u_{n-2} and u_{n-1}, and after u_{n-1}. (Indeed, $|S_n| = n! = n \cdot (n-1)! = n|S_{n-1}|$.)

How does this insertion affect the number of descents? There are two cases.

1. Suppose we insert n into an "ascent slot" of u, i.e. between u_i and u_{i+1} where $u_i < u_{i+1}$. Then we create a new descent (from n to u_{i+1}), so $\operatorname{des}(w) = \operatorname{des}(u) + 1$ is this case.
2. Suppose we insert n into a "descent slot" (defined analogously). Then we replace a descent $u_i \to u_{i+1}$ with a descent $n \to u_{i+1}$, so the number of descents does not change.

We can see that the slot before u_1 acts as an ascent slot, and the slot after u_{n-1} acts as a descent slot (you can imagine that $u_0, u_n = -\infty$).

Assume that $\operatorname{des}(w) = k$. If w falls into the first case above, $\operatorname{des}(u) = k-1$, and u has $n-k$ ascent slots (including before u_1). So this case contributes $(n-k)\tilde{A}(n-1,k-1)$.

Similarly, case 2 requires $\operatorname{des}(u) = k$ and n to be inserted into one of $k-1$ descent slots (including after u_{n-1}), so this yields $(k-1)A(n-1,k)$ possibilities. Adding these cases up yields our desired recurrence, and we can check that the initial conditions are correct, so $A(n,k)$ indeed counts permutations in S_n with k descents and we are done.

Stirling numbers

There are a two kinds of Stirling numbers, and the first kind can be either unsigned or signed.

- We define **signless Stirling numbers of the first kind** as

$$\begin{aligned} c(n,k) &:= \#\{w \in S_n \mid \operatorname{cyc}(w) = k\} \\ &= \#\{w \in S_n \mid \operatorname{rec}(w) = k\} \end{aligned}$$

  (we proved the second equality earlier).
- Often, **signed Stirling numbers of the first kind** are helpful.

$$s(n,k) := (-1)^{n-k} c(n,k).$$

- **Stirling numbers of the second kind** $S(n,k)$ count the number of "set partitions" of $[n]$ with k blocks.

What is a set partition?

Definition (Set partitions).

A **set partition** $\pi = (B_1 \mid B_2 \mid \cdots \mid B_k)$ of $[n] := \{1, 2, \ldots, n\}$ satisfies:

- $[n] = B_1 \sqcup B_2 \sqcup \cdots \sqcup B_k$ (disjoint union)
- $B_i \neq \varnothing$

Also, the order of blocks does not matter.

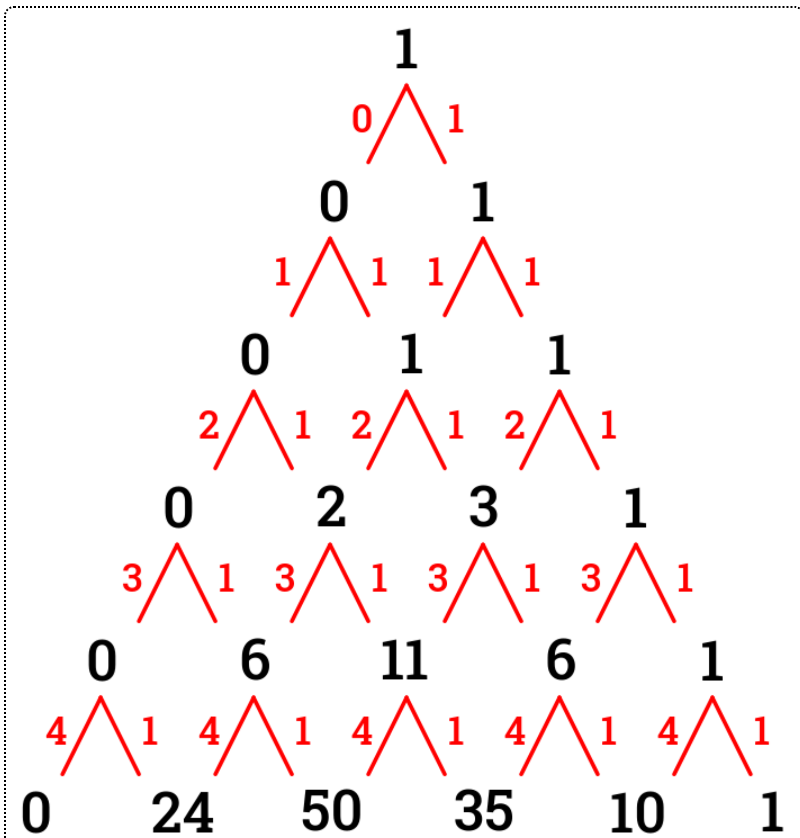
For example, $S(4,2) = 7$: the set partitions are

$$\begin{gathered} (1\ 2\ 3 \mid 4), (1\ 2\ 4 \mid 3), (1\ 3\ 4 \mid 2), (2\ 3\ 4 \mid 1), \\ (1\ 2 \mid 3\ 4), (1\ 3 \mid 2\ 4), (1\ 4 \mid 2\ 3) \end{gathered}$$

Similar to the Eulerian triangle, we can form some Stirling triangles (for both the first and the second kind).

Here's the first Stirling triangle:

$$\begin{array}{ccccccc} & & & c(0,0) & & & \\ & & c(1,0) & & c(1,1) & & \\ & c(2,0) & & c(2,1) & & c(2,2) & \\ \iddots & & & \vdots & & & \ddots \end{array}$$



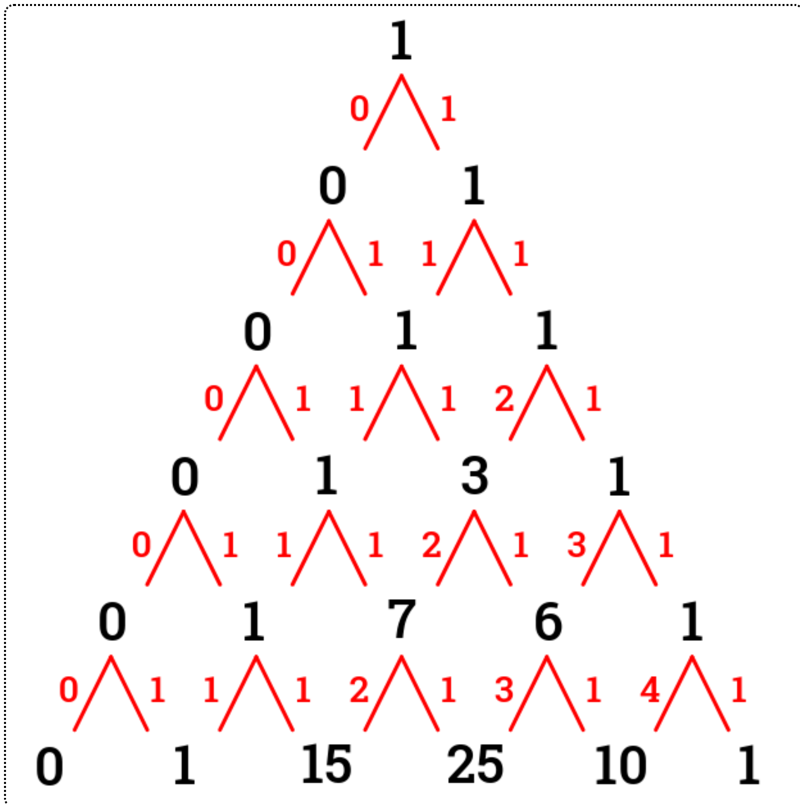
(Remember that we're counting permutations with a given number of cycles here, since these are Stirling numbers of the first kind.)

We can observe that $c(n,n) = 1$ and $c(n,1) = n!$. Once again, we have a Pascal-like weighted recurrence relation, namely

$$c(n,k) = c(n-1,k-1) + \underline{(n-1)} \cdot c(n-1,k)$$

Here's the second Stirling triangle, which also reveals a recurrence relation:

$$\begin{array}{ccccccc} & & & S(0,0) & & & \\ & & S(1,0) & & S(1,1) & & \\ & S(2,0) & & S(2,1) & & S(2,2) & \\ \iddots & & & \vdots & & & \ddots \end{array}$$



$$S(n,k) = S(n-1,k-1) + \underline{k} \cdot S(n-1,k)$$

Our proofs of these recurrences will be similar to our proof for the Eulerian numbers (by thinking about inserting n into our permutations or partitions).

Proof for the first kind. This time, write our permutations in cycle notation. Suppose we have $u = (\ldots)(\ldots)\cdots(\ldots) \in S_{n-1}$ and want to insert n somewhere to obtain $w \in S_n$. There are two cases:

- either it becomes a fixed point (only one possibility; $\mathrm{cyc}(w) = \mathrm{cyc}(u) + 1$),
- or we insert it into an existing cycle ($n-1$ possibilities since n can go after any number; $\mathrm{cyc}(w) = \mathrm{cyc}(u) + 1$).
  This yields our desired recurrence.

Proof for the second kind. Suppose $\delta = (B_1 \mid B_2 \mid B_3 \mid \cdots \mid B_k)$ is a set partition of $[n-1]$. If we want to extend this to a set partition of $[n]$, once again we can either make it a singleton (increasing the number of blocks by 1) or adding it to an existing block (for which we have k cases). Similarly, this proves our recurrence.

Alright, so we have these two kinds of Stirling numbers. Why are they related?

We'll need some more notation.

Definition (Falling factorial / Pochhammer symbol).

We denote the **falling factorial**, also known as the **Pochhammer symbol** $(x)_n$, as follows:

$$(x)_n := x(x-1)(x-2)\dots(x-n+1).$$

Now, we have the following:

Theorem 9 (Falling factorials and Stirling numbers).

$$\begin{aligned} (1) \qquad & \sum_{k=0}^{n} s(n,k)x^k = (x)_n \\ (2) \qquad & \sum_{k=0}^{n} S(n,k)(x)_k = x^n. \end{aligned}$$

(Recall that $s(n,k)$ are like $c(n,k)$, but with alternating signs.)

For example, if $n = 3$, we have:

$$\begin{aligned} (1) \qquad & -0x^0 + 2x^1 - 3x^2 + x^3 = x(x-1)(x-2) \\ (2) \qquad & 0 \cdot 1 + 1 \cdot x + 3 \cdot x(x-1) + 1 \cdot x(x-1)(x-2) = x^3. \end{aligned}$$

We can also view this relation using linear algebra. Our two types of Stirling numbers allow us to convert between these two bases of the linear space $\mathbb{R}[x]$:

- $1, x, x^2, x^3, \dots$
- $1, x, x(x-1), x(x-1)(x-2), \dots$

So we can use Stirling numbers to create change-of-basis matrices. In fact,

> **Corollary** (Matrices of Stirling numbers).
>
> Fix some N (the maximal polynomial degree we consider). Then
>
> $$[s(n,k)]_{1\le k,n\le N} = [S(n,k)]^{-1}_{1\le k,n\le N}.$$

There are multiple ways to prove Theorem 9. The boring way is to use induction and recurrence relations (not terribly hard; you can do it as an exercise), but there are some interesting ways to proving it bijectively/combinatorially. We will go over them next week.

Lecture 10 (February 24)

You should start the problem set!

Relating Stirling numbers and falling factorials

We will combinatorially prove

$$\sum_{k=0}^{n} S(n,k)(x)_k = x^n.$$

It suffices to show that the two polynomials coincide at infinitely many points, since that would mean that they are identically equal. So we can assume $x \in \mathbb{Z}_{\ge 0}$.

We want to combinatorially interpret both sides. The right hand side, x^n, is the number of all functions $f\colon \{1,2,\ldots,n\} \to \{1,2,\ldots,x\}$, since you have x choices for each $f(k)$.

In order for this to equal the left hand side, we want to interpret each term $S(n,k)(x)_k$ as some subset of the functions. We can rewrite this term as $k!S(n,k)\binom{n}{k}$. We look at each factor individually:

- $\binom{x}{k}$ is the number of k-element subsets $\{i_1, i_2, \ldots, i_k\}$ in $[x]$
- $S(n,k)$ is the number of set partitions of $[n]$ into k (unordered) blocks

- $k!$ can be thought of as the number of ways to order those blocks.

So, $k!S(n,k)$ is the number of surjective maps $f\colon [n] \to \{i_1, \ldots, i_k\}$, since for any set partition with ordered blocks, we can make the first block map to i_1, the second block map to i_2, etc. In other words, the bth block is the preimage of i_b in the function.

As a result, $k!S(n,k)\binom{x}{k}$ is the number of functions $f\colon [n] \to [x]$ with a k-element preimage. Clearly, adding this up for all possible values of k will yield the total number of functions x^n as desired. (Note that the $k = 0$ case yields 0.)

Since these two polynomials have finite degree n and are equal at infinitely many points (every nonnegative integer), they must be exactly the same, and we are done.

The notes for this lecture continue in the next file (Notes 04. Partially Ordered Sets).